

Random variables

- Non-Gaussian distributions

Statistical notation

Shorthands like these

$$W \sim P_\theta$$

$$W|\theta \sim P_\theta$$

indicate an **uncertain quantity** and its distribution which depends upon certain or uncertain parameters

A **probability distribution** is a shorthand for

- what values our uncertain quantity may attain
- how to compute probabilities associated with these values

Distributions of **parametric quantities**, ie random variables, are characterized by **probability functions**

- A random variable may be **discrete** and it is associated with a *probability mass function*
- A random variable may be **continuous** and it is associated with a *probability density function*

Discrete random variables

Bernoulli

A Bernoulli random variable is denoted with

$$w \sim \text{Bernoulli}(\pi)$$

This variable attains integer values in $\{0, 1\}$

The probability mass function is given by

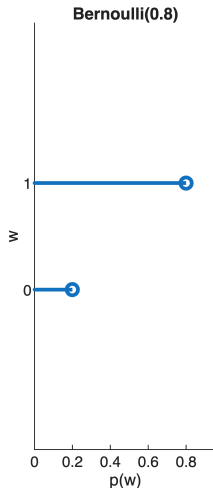
$$\begin{aligned} p(w) = \text{Bernoulli}(w; \pi) &= \begin{cases} \pi, & \text{for } w = 1 \\ 1 - \pi, & \text{for } w = 0 \end{cases} \\ &= \pi^w (1 - \pi)^{1-w} \end{aligned}$$

The parameter is real $\pi \in [0, 1]$

- It **matters** because it is a simple model for many systems
- It is **simulated** by

$$w = (\text{rand} < \pi);$$

which reads $w = 1_{u < \pi}$ and $u \sim \text{Uniform}_{[0,1]}$



Binomial

A binomial random variable is denoted with

$$w \sim \text{Binomial}(J, \pi)$$

This variable attains integer values in $\{0, 1, \dots, J\}$

The probability mass function is given by

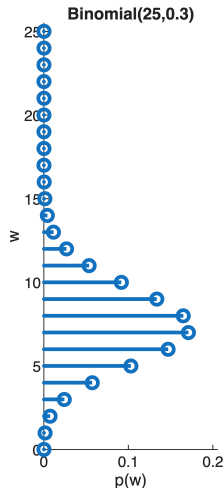
$$p(w) = \text{Binomial}(w; J, \pi) = \binom{J}{w} \pi^w (1 - \pi)^{J-w}$$

The parameters are integer $J = 0, 1, \dots$ and real $\pi \in [0, 1]$

- It **matters** because it is a simple model well suited for very many systems
- It is **simulated** by

$$w = \text{sum}(\text{rand}(J, 1) < \pi);$$

which reads $w = \sum_{j=1}^J \beta_j$ and $\beta_j \sim \text{Bernoulli}(\pi)$



Poisson

A Poisson random variable is denoted with

$$w \sim \text{Poisson}(\mu)$$

This variable attains integer values in $\{0, 1, \dots\}$

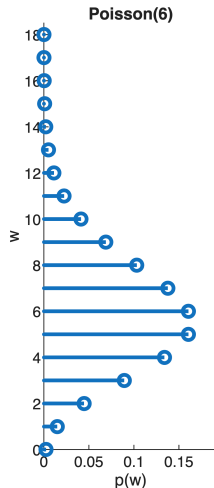
The probability mass function is given by

$$p(w) = \text{Poisson}(w; \mu) = \frac{\mu^w}{w!} e^{-\mu}$$

The parameter is real $\mu \in (0, \infty)$

- It **matters** because it lets us calculate the probability of a specific number of random, independent events happening over a fixed block of time or space.
- It is approximated by

$$\text{Poisson}(\mu) \approx \text{Binomial}(J, \mu/J), \quad J \gg 1$$



Geometric

A geometric random variable is denoted with

$$w \sim \text{Geometric}(\pi)$$

This variable attains integer values in $\{1, 2, \dots\}$

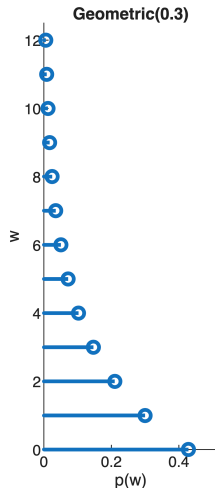
The probability mass function is given by

$$p(w) = \text{Geometric}(w; \pi) = (1 - \pi)^{w-1} \pi$$

The parameter is real $\pi \in (0, 1]$

- It **matters** because it predicts the number of trials needed to achieve a single success in a series of independent events
- It **matters** also because it is characterized by memorylessness

$$\text{Prob}(w > s + t | w > s) = \text{Prob}(w > t)$$



Continuous random variables

Uniform

A uniform random variable is denoted with

$$w \sim \text{Uniform}_{[\alpha, \beta]}$$

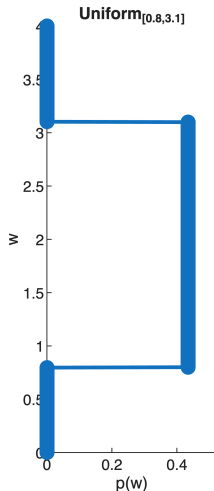
This variable attains real values in $[\alpha, \beta]$

The probability density function is given by

$$p(w) = \begin{cases} \frac{1}{\beta - \alpha}, & \text{for } w \in [\alpha, \beta] \\ 0, & \text{for } w \notin [\alpha, \beta] \end{cases}$$

The parameters are real $-\infty < \alpha < \beta < +\infty$

- It matters because it is the only random variable that can be simulated in a computer



Beta

A beta random variable is denoted with

$$w \sim \text{Beta}(\alpha, \beta)$$

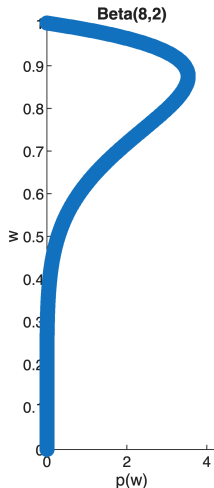
This variable attains real values in $[0, 1]$

The probability density function is given by

$$p(w) = \begin{cases} \frac{w^{\alpha-1}(1-w)^{\beta-1}}{B(\alpha, \beta)}, & \text{for } w \in [0, 1] \\ 0, & \text{for } w \notin [0, 1] \end{cases}$$

The parameters are reals $\alpha, \beta \in (0, \infty)$

- It **matters** because it is a convenient prior for Bernoulli likelihoods in the natural parametrization



von Mises

A von Mises random variable is denoted with

$$w \sim \text{vonMises}(\mu, \kappa)$$

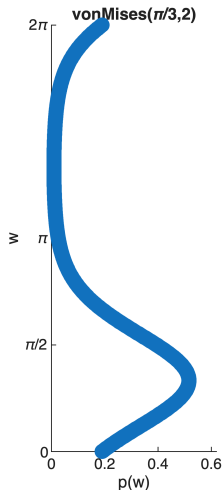
This variable attains real values in $[0, 2\pi]$

The probability density function is given by

$$p(w) = \begin{cases} \frac{e^{\kappa \cos(w-\mu)}}{2\pi I_0(\kappa)}, & \text{for } w \in [0, 2\pi] \\ 0, & \text{for } w \notin [0, 2\pi] \end{cases}$$

The parameters are reals $\mu \in (-\infty, +\infty)$ and $\kappa \in (0, \infty)$

- It **matters** because it is the circular analogue to the normal distribution for directional or angular data



Exponential

An exponential random variable is denoted with

$$w \sim \text{Exponential}(\mu)$$

This variable attains real values in $[0, \infty)$

The probability density function is given by

$$p(w) = \begin{cases} \frac{1}{\mu} e^{-w/\mu}, & \text{for } w \geq 0 \\ 0, & \text{for } w < 0 \end{cases}$$

The parameter is real $\mu \in (0, \infty)$

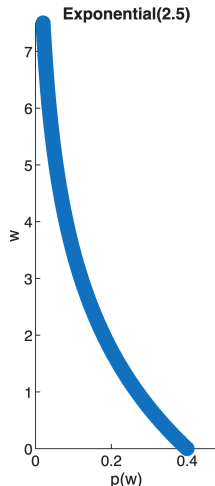
- It matters because it is characterized by memorylessness

$$\text{Prob}(w > s + t | w > s) = \text{Prob}(w > t)$$

- It is simulated by

$$w = -\mu * \log(\text{rand});$$

which reads $w = -\mu \log(u)$ and $u \sim \text{Uniform}_{[0,1]}$



Gamma

A gamma random variable is denoted with

$$w \sim \text{Gamma}(\phi, \psi)$$

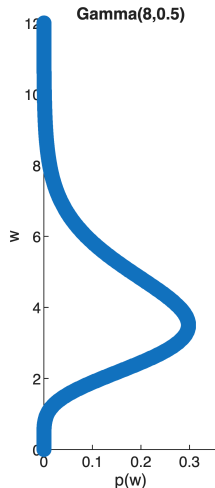
This variable attains real values in $[0, \infty)$

The probability density function is given by

$$p(w) = \begin{cases} \frac{1}{\Gamma(\phi)\psi^\phi} w^{\phi-1} e^{-w/\psi}, & \text{for } w \geq 0 \\ 0, & \text{for } w < 0 \end{cases}$$

The parameters are real $\phi, \psi \in (0, \infty)$

- It matters because it generalizes exponential random variables
- It matters because it is a convenient prior for very many likelihoods



Weibull

A Weibull random variable is denoted with

$$w \sim \text{Weibull}(\phi, \psi)$$

This variable attains real values in $[0, \infty)$

The probability density function is given by

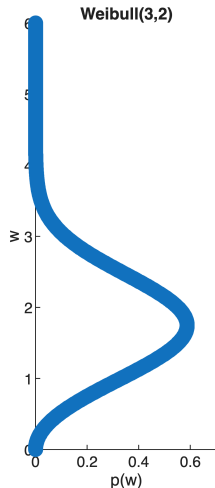
$$p(w) = \begin{cases} \frac{\phi}{\psi} \left(\frac{w}{\psi} \right)^{\phi-1} e^{-(w/\psi)^\phi}, & \text{for } w \geq 0 \\ 0, & \text{for } w < 0 \end{cases}$$

The parameters are real $\phi, \psi \in (0, \infty)$

- It is simulated by

$$w = \psi * (-\log(\text{rand}))^{1/\phi};$$

which reads $w = \psi(-\log(u))^{1/\phi}$ and $u \sim \text{Uniform}_{[0,1]}$



Beta-prime

A beta-prime random variable is denoted with

$$w \sim \text{BetaPrime}(\alpha, \beta)$$

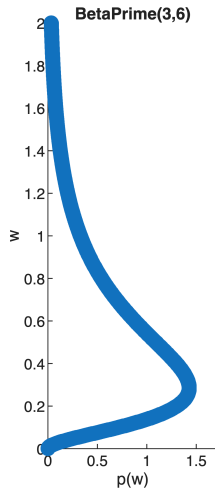
This variable attains real values in $[0, \infty)$

The probability density function is given by

$$p(w) = \begin{cases} \frac{w^{\alpha-1}(1+w)^{-\alpha-\beta}}{B(\alpha, \beta)}, & \text{for } w \geq 0 \\ 0, & \text{for } w < 0 \end{cases}$$

The parameters are reals $\alpha, \beta \in (0, \infty)$

- It matters because it is a convenient prior for Bernoulli likelihoods parametrized in terms of odds



χ

A χ random variable is denoted with

$$w \sim \chi_k$$

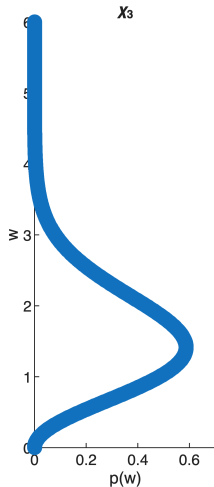
This variable attains real values in $[0, \infty)$

The probability density function is given by

$$p(w) = \begin{cases} \frac{w^{k-1} e^{-w^2/2}}{2^{k/2-1} \Gamma(k/2)}, & \text{for } w \geq 0 \\ 0, & \text{for } w < 0 \end{cases}$$

The parameter is real $k \in (0, \infty)$

- It matters because it is distribution of the Euclidean norm of multivariate normals



χ^2

A χ^2 random variable is denoted with

$$w \sim \chi_k^2$$

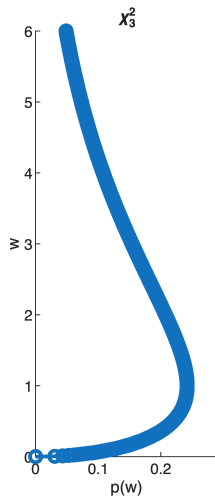
This variable attains real values in $[0, \infty)$

The probability density function is given by

$$p(w) = \begin{cases} \frac{w^{k/2-1} e^{-w/2}}{2^{k/2} \Gamma(k/2)}, & \text{for } w \geq 0 \\ 0, & \text{for } w < 0 \end{cases}$$

The parameter is real $k \in (0, \infty)$

- It **matters** because it is distribution of the squared Euclidean norm of multivariate normals



Student-t

A Student-t random variable is denoted with

$$w \sim \text{StudentT}_\nu(\mu, \nu)$$

This variable attains real values in $(-\infty, +\infty)$

The probability density function is given by

$$p(w) = \frac{\Gamma(\frac{\nu+1}{2})}{\Gamma(\frac{\nu}{2}) \sqrt{\nu\pi\nu}} \left(1 + \frac{1}{\nu} \frac{(w - \mu)^2}{\nu}\right)^{-(\nu+1)/2}$$

The parameters are reals $\nu, \nu \in (0, \infty)$ and $\mu \in (-\infty, +\infty)$

- It **matters** because it has heavy tails
- It is approximated by

$$\text{Normal}(\mu, \sigma^2) \approx \text{StudentT}_\nu(\mu, \sigma^2), \quad \nu \gg 1$$

